

When Agreement Fails: Visual Anchoring and Multimodal Pseudo-Label Reliability in Classroom Behavior Recognition

Yan Ma^{1,3}, Lizhuo Zhang^{2*}

1 School of Education, Hunan Agricultural University, Changsha, China

2 School of Information and Intelligent Science and Technology, Hunan Agricultural University, Changsha, China

3 School of Foreign Studies, Changsha University of Science and Technology, Changsha, China

* zhanglizhuo@hunau.edu.cn

Abstract

Agreement-based pseudo-label filtering is often treated as a simple reliability heuristic for multimodal large language models (MLLMs), but its actual value for downstream adaptation is unclear. This study tests that assumption on three publicly available SCB sub-datasets for classroom behavior recognition. The main annotation pipeline uses two MLLMs (Qwen2-VL-7B-Instruct and LLaVA-1.5-7B), and a five-model robustness check tests whether the central conclusion depends on that pair. The analysis combines annotation-quality measurement, downstream CLIP ViT-L/14 fine-tuning, and mechanism-oriented diagnostics.

Agreement filtering is not a reliable default in this setting. On BowTurnHead, the agreement-retained linear-probe result is $15.53\% \pm 1.00\%$ (repeated-seed mean), whereas a fixed-seed retention diagnostic at 25% random retention reaches 85.96%; these values come from different protocols and are reported as complementary mechanism checks. Retention-ratio analysis identifies three failure modes: data-volume collapse, distributional bias, and quality ceiling.

Visual anchoring is then used as a mechanism-oriented explanation for when agreement fails. Protocol-matched per-class zero-shot CLIP accuracy serves as an anchoring proxy and organises category-level reliability differences. Because the selective-routing diagnostic reuses the same zero-shot branch for low-anchor classes, Scheme B is interpreted as an upper-bound diagnostic rather than independent confirmatory evidence. Under oracle routing, Scheme B reaches 58.37% overall on TeacherBehavior.

Across five tested MLLMs, no single annotator is uniformly best across datasets. The main linear-probe matrix uses repeated seeds, and a targeted five-seed LoRA replication (43–47) on BowTurnHead and TeacherBehavior shows lower cross-seed variability under GT than under pseudo-label supervision (mean SD = 0.107% vs. 0.495%), consistent with label quality as the primary constraint on attainable performance. Remaining LoRA and auxiliary diagnostics are retained as protocol-specific reference analyses. These conclusions are bounded to the SCB benchmark and single-frame classroom imagery; broader cross-dataset validation is required.

Introduction

Automatic recognition of classroom behaviors—teacher instructional actions, student engagement postures, and off-task micro-movements—is a prerequisite for scalable educational quality assessment and intelligent classroom analytics [1, 2]. Supervised deep learning models have demonstrated strong recognition performance on this task [3, 4], but they require large quantities of labeled images that are labor-intensive to collect: a trained annotator must watch each video clip, identify every person in the frame, draw bounding boxes, and assign the correct activity label, often under ambiguous multi-label conditions. This annotation bottleneck severely limits the scale and diversity of available classroom datasets, and motivates the search for lower-cost labeling alternatives.

Multimodal large language models (MLLMs) such as Qwen [5] and LLaVA [6] have recently demonstrated impressive visual understanding across a wide range of recognition tasks. Given an image and a list of candidate categories, an MLLM can produce a category prediction without any task-specific training, making them practical candidates for automatic annotation pipelines. Several recent studies have explored LLM-assisted annotation in text-heavy domains [7, 8], and the LATTECLIP framework [9] has shown that MLLM-synthesized text descriptions can drive unsupervised CLIP fine-tuning in general visual domains. However, whether MLLMs can serve as reliable annotators for *fine-grained, domain-specific* behaviors—particularly those requiring interpretation of pedagogical intent rather than surface visual features—has not been systematically studied.

Classroom activity recognition poses distinctive challenges that make this question non-trivial. Some behaviors have clear visual signatures: a teacher writing on a blackboard produces a characteristic image that any vision model should recognise. Others are intent-dependent: the visual difference between a teacher “answering a student’s question” and a teacher “giving a lecture” may be imperceptible from a single frame. It is therefore plausible that MLLM annotation quality varies substantially across categories, and that a simple pipeline of “annotate everything, then fine-tune” may be undermined by a small number of systematically mis-labelled classes. This paper refers to that category-specific degree of visual identifiability as *visual anchoring*.

From an engineering perspective, the practical question is not whether MLLMs can produce pseudo-labels at all, but when those labels are reliable enough to improve downstream adaptation and when filtering removes the very supervision the model needs.

This study therefore addresses one central question: *when can MLLM-generated pseudo-labels provide a reliable training signal for classroom behavior recognition, and when does agreement filtering make that signal worse rather than better?*

Answering that question requires three linked tests rather than three co-equal goals. First, the paper measures category-level annotation reliability to determine where single-frame classroom crops are visually anchored enough for automatic labeling. Second, it tests whether filtered and unfiltered pseudo-label sets improve downstream CLIP adaptation relative to zero-shot inference and a ground-truth reference. Third, it uses retention controls, anchor-aware selective routing, and a five-model robustness check to distinguish between data-volume loss, sampling bias, category semantics, and simple dependence on the original annotator pair.

The main pseudo-label annotation pipeline uses two MLLMs (Qwen2-VL-7B-Instruct and LLaVA-1.5-7B) to annotate cropped bounding-box regions from three SCB sub-datasets covering 13 behavior categories across more than 87,000 bounding boxes. A later five-model robustness analysis is treated as a boundary condition check rather than as the paper’s main claim. The core answer therefore comes from the downstream fine-tuning matrix, while the additional diagnostics explain why that answer changes across datasets and categories.

The contributions are therefore fourfold.

- (1) **Pseudo-label usefulness is category-conditional rather than global.** Visual anchoring separates classes that MLLMs can label reliably from classes that remain structurally ambiguous under single-frame crops.
- (2) **Agreement filtering is not a safe denoising default.** The paper shows that agreement filtering can fail through data-volume collapse, distributional bias, and shared label error, even when it appears

to improve retained-subset annotation accuracy.

- (3) **Anchor-aware selective routing is useful only as a bounded diagnostic.** An oracle selective-routing analysis shows where pseudo-label supervision remains useful when restricted to structurally reliable classes, while also making clear that this result is an upper bound rather than a deployable policy.
- (4) **The main limitation is label quality and category semantics, not only optimizer noise or a single annotator choice.** Repeated-seed downstream evaluation, targeted LoRA replications, and a five-model robustness check all point to the same boundary condition in the present setting.

Related Work

Classroom Activity Recognition

Automated analysis of classroom teaching and learning behaviors has been approached through video [10], audio [11], and multimodal sensing [12]. CNN-based methods achieved early success on student engagement detection [4], and transformer architectures have more recently enabled richer spatiotemporal modelling [3]. A persistent challenge across all these works is the scarcity of labeled data: classroom recordings raise privacy concerns, annotation requires domain expertise, and activity categories are often defined by pedagogical intent that is not directly visible in individual frames. This study directly addresses this bottleneck by evaluating whether MLLMs can substitute for human annotators, thereby reducing labeling cost.

Vision–Language Models and Zero-Shot Transfer

CLIP [13] established contrastive image–text pretraining as a powerful paradigm for zero-shot visual recognition, enabling classification by matching image embeddings to natural language category descriptions. Subsequent work has extended this framework through improved training data [14, 15], alternative loss functions [16], masked image modeling [17], and data filtering networks [18]. Domain-specific applications include medical imaging [19], remote sensing [20], and action recognition [21]. Zero-shot CLIP performance is known to be sensitive to prompt design [13], and methods such as CoOp [22] and CoCoOp [23] learn soft prompt embeddings to adapt CLIP to target domains with minimal supervision. In the classroom domain, however, zero-shot CLIP performance across multiple scene types and model variants remains understudied.

LLM- and MLLM-Assisted Data Annotation

Using language models to generate or refine training labels has attracted growing interest. Early work showed that GPT-3/4-class models can produce competitive NLP annotations for tasks such as stance classification and named entity recognition [7, 8]. In the multimodal setting, Zhang et al. [24] benchmark zero-shot MLLM annotation for facial expression images, finding large variance across model families and expression categories—a pattern that parallels the results reported here. However, that study stops at annotation benchmarking; it does not test downstream CLIP adaptation, compare filtering protocols, or analyse why particular categories fail. Most relevant to this study is LATTECLIP [9], which uses MLLM-generated rich textual descriptions to fine-tune CLIP without any human labels in a general visual setting; the authors note that annotation quality is especially low for classes that are “not visually discriminative,” foreshadowing the visual anchoring finding. Recent open-weight MLLM progress in 2024–2025 has also changed the practical annotation landscape. Model families such as LLaVA-NeXT, Qwen2.5-VL, and InternVL2 report stronger general visual reasoning and instruction following capabilities [25–27], while broader multimodal benchmarks such as MMMU still show that domain-specific reasoning remains uneven and task dependent [28]. This is directly relevant to pseudo-label pipelines because better average MLLM capability does not automatically imply reliable category coverage under constrained prompts and single-frame evidence. This study differs from LATTECLIP in three ways: it operates in a fine-grained domain-specific setting, uses bounding-box-level rather than image-level annotations, and explicitly characterises the relationship between per-category annotation accuracy and downstream

fine-tuning performance. More broadly, the present study combines three threads that are usually treated separately: MLLM-based automatic annotation, pseudo-label filtering, and classroom behavior recognition. To the best of current knowledge, no prior work has systematically evaluated MLLM annotation reliability for classroom activity recognition or provided per-category diagnostic evidence of annotation failure modes.

Pseudo-Label Filtering Strategies

Filtering noisy pseudo-labels is a standard component of semi-supervised and weakly supervised learning pipelines [29,30]. Common strategies include confidence thresholding, consistency regularization across augmentations, and ensemble agreement [31]. In the classic semi-supervised setting, however, consistency filtering usually compares predictions from the same model under perturbation or augmentation, whereas the present study examines agreement between two different MLLMs on the same crop. That distinction matters because cross-model agreement can remove not only noisy samples but also large, systematically informative regions of the training distribution when the two annotators fail in different ways. The agreement-based filter used in this study—retaining samples where two independently queried MLLMs assign the same label—is conceptually related to multi-annotator agreement in crowdsourcing [32], where consensus labels are generally higher quality but fewer in number. Another major family of pseudo-label filtering methods uses calibrated confidence or uncertainty estimates rather than agreement alone. That comparison is relevant here, but the executed annotation protocol uses index-only prompting and therefore does not expose a calibrated scalar confidence signal; for that reason, the present experiments test only the screening cues actually produced by the annotation pipeline, namely valid-output screening and cross-model agreement. A key finding of this study is that, in the observed retention regime (20–67% on the training splits), the data-volume cost and sampling bias of agreement filtering can overwhelm any quality benefit, leading to substantial over-fitting or negligible downstream gains. In these experiments, this pattern directly contradicts the common intuition that cross-model agreement automatically yields better training data. This complements findings in the NLP annotation literature showing that LLM label quality does not monotonically improve with filtering stringency when the underlying model systematically errs on certain input types [8]. Related 2024 reliability-oriented evaluations in multimodal settings similarly suggest that robust deployment requires explicit uncertainty or error-structure checks rather than assuming that stricter filtering always improves supervision quality [24,28]. Taken together, the gap in prior work is not the absence of MLLM annotation studies, pseudo-label filtering methods, or classroom recognition benchmarks in isolation, but the lack of a single study that tests these components jointly and explains when and why the combined pipeline fails in a classroom-behavior setting.

Datasets and Experimental Setup

The study is organised as a top-down evaluation pipeline aligned with the three research questions stated in the Introduction. It first measures bbox-level pseudo-label reliability under a common MLLM annotation protocol, then tests whether filtered or unfiltered pseudo-labels improve downstream CLIP fine-tuning relative to a training-free zero-shot reference, and finally uses targeted diagnostic analyses to explain the observed gains and failure modes. These linked analyses include category-level anchoring diagnostics, retention-ratio tests for data-volume effects, a category-aware routing diagnostic on TeacherBehavior, and a cross-model robustness check to determine whether the main annotation patterns persist under model substitution.

Datasets

All experiments use three sub-datasets from the publicly available SCB (Smart Classroom Behavior) benchmark [33], which provides bounding-box annotations for classroom activity images captured in real Chinese K–12 classrooms. The three sub-datasets cover complementary scene types and difficulty levels, summarised in Table 1.

Table 1. Overview of the three SCB sub-datasets used in this study. “BB” = bounding boxes.

Sub-dataset	Classes	Train images	Val images	Total BB (train/val)
BowTurnHead	2	1,905	505	12,365 / 3,753
HandriseReadWrite	3	5,193	1,671	34,524 / 12,848
TeacherBehavior	8	8,984	3,240	40,372 / 15,327

Class balance is uneven in all three sub-datasets and is therefore reported explicitly in Table 2. BowTurnHead is skewed toward *turn-head*; HandriseReadWrite is dominated by *read*; and TeacherBehavior is dominated by *stand*, *teacher*, and *blackboard*, while *on-stage interaction* and *blackboard-writing* are rare. These imbalances provide relevant context for interpreting agreement retention bias and selective-routing behavior.

Table 2. Per-class bounding-box counts for the SCB sub-datasets. These counts expose the class imbalance that interacts with agreement filtering and anchor-aware routing.

Sub-dataset	Class	Train BB	Val BB
BowTurnHead	bow-head	4,422	540
BowTurnHead	turn-head	7,943	3,213
HandriseReadWrite	hand-raise	10,538	2,915
HandriseReadWrite	read	17,539	6,539
HandriseReadWrite	write	6,447	3,394
TeacherBehavior	guide	1,155	449
TeacherBehavior	answer	2,574	853
TeacherBehavior	on-stage interaction	528	149
TeacherBehavior	blackboard-writing	821	277
TeacherBehavior	teacher	8,490	3,228
TeacherBehavior	stand	13,932	4,967
TeacherBehavior	screen	5,025	1,959
TeacherBehavior	blackboard	7,847	3,445

BowTurnHead contains images of students exhibiting two micro-action classes: *bow-head* and *turn-head*. The *bow-head* class typically corresponds to a downward head posture associated with distraction or phone use. The binary nature of the task and the visually distinct postures make this the easiest sub-dataset.

HandriseReadWrite contains three student behavior classes: *hand-raise*, *read*, and *write*. These classes require observing upper-body posture and hand position, introducing moderate visual ambiguity between reading and writing.

TeacherBehavior contains eight teacher activity classes: guide, answer, on-stage interaction, blackboard-writing, teacher (general presence), stand, screen, and blackboard. The large class count, overlapping visual appearances, and the prevalence of intent-dependent categories (e.g., “answer” versus “guide” are visually nearly identical) make this the most challenging sub-dataset.

Throughout the manuscript, class names follow the index order used in the annotation and evaluation scripts: BowTurnHead = {0: *bow-head*, 1: *turn-head*}; HandriseReadWrite = {0: *hand-raise*, 1: *read*, 2: *write*}; TeacherBehavior = {0: *guide*, 1: *answer*, 2: *on-stage interaction*, 3: *blackboard-writing*, 4: *teacher*, 5: *stand*, 6: *screen*, 7: *blackboard*}.

All annotations are provided in YOLO format (class index, normalised centre coordinates, width, height). Each bounding box defines a single activity instance; images may contain multiple bounding boxes corresponding to different individuals or simultaneous activities.

Ethics and Data Governance

This study is a secondary analysis of a previously released public benchmark of classroom images and bounding-box annotations. The authors did not recruit participants, collect new data, intervene in classroom

activities, or access metadata beyond the released image and label files. The analysis uses only the released benchmark files, reports aggregate results, and does not reproduce identifiable classroom images. Under this secondary-analysis design, additional institutional review and informed consent were not required for the present study. Ethical oversight for the original classroom data collection remains with the SCB dataset creators and the institutions that approved its release. The present work relies on those original approvals and follows the authors’ institutional policy for secondary analysis of public child image datasets: no re-identification attempts, no release of new identifiable child images, and aggregate-only reporting.

MLLM Annotation Pipeline

Bounding Box Cropping

For each annotated bounding box, the corresponding image region is cropped with a 5% margin on each side to provide contextual information around the target. Crops smaller than 32×32 pixels after margin expansion are discarded. This yields 87,261 training crops and 31,928 validation crops across the three sub-datasets.

Models

In the main pseudo-label annotation pipeline, two locally deployable MLLMs serve as annotators: **Qwen2-VL-7B-Instruct** [5] (hereafter Qwen) and **LLaVA-1.5-7B** [6] (hereafter LLaVA), both running on dedicated GPUs to enable parallel annotation. The study selects locally deployable 7B-scale models for three reasons: reproducibility without API dependency, cost-free inference at scale, and comparability between the two models under identical hardware conditions.

Prompt Design

Both models receive a uniform zero-shot prompt listing the category names with their indices, followed by the instruction to respond with the index only:

You are a classroom behavior recognition expert. Given one cropped classroom image, choose the single best matching class. Reply with only the class index number. No explanation. Classes: [0: class_0, 1: class_1, ...]. Answer with index only:

The same prompt serves all three sub-datasets; only the class list changes. No dataset-specific synonyms, textual definitions, or auxiliary class descriptions are added beyond the class names shown in the prompt. This index-only format standardises the output space across models and reduces parsing ambiguity, but it does not measure uncertainty and may force low-confidence decisions into a single discrete label. For all executed annotation runs, decoding is deterministic (`do_sample=False`), and the response length is capped at `max_new_tokens=8`.

Table 3 summarises the label-index order used in the executed runs. The same dataset-specific class list is passed to both MLLMs from a shared configuration in `step1_11m_annotate.py`, and downstream evaluation reuses the same order for label decoding. Valid model outputs are single in-range integers such as “0”, “1”, or “4”; any non-integer or out-of-range string is recorded as invalid rather than coerced to a class label.

Table 3. Label-index order audit for the index-only prompt. The same ordering is used for both MLLM annotation and downstream label decoding, providing a direct consistency check against index-mapping errors.

Dataset	Index order used in prompt and evaluation
TeacherBehavior	0: guide; 1: answer; 2: on-stage interaction; 3: blackboard-writing; 4: teacher; 5: stand; 6: screen; 7: blackboard
HandriseReadWrite	0: hand-raise; 1: read; 2: write
BowTurnHead	0: bow-head; 1: turn-head

Fig 1 summarizes the full experimental pipeline, from bbox cropping and MLLM annotation to pseudo-label filtering, CLIP fine-tuning, and diagnostic interpretation.

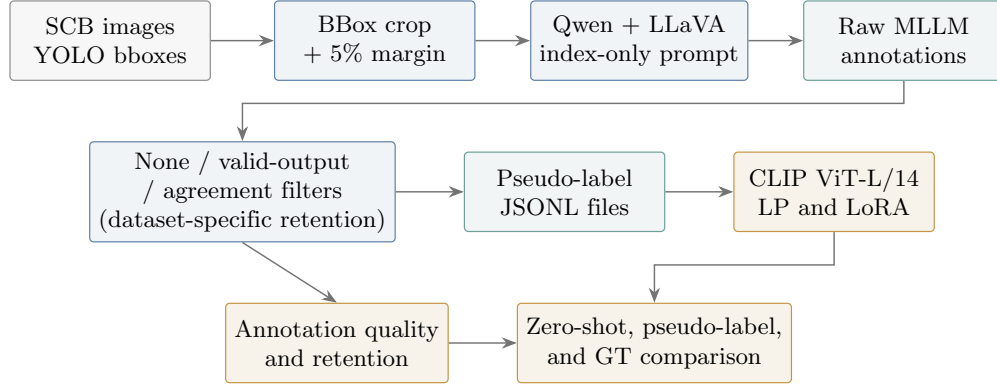


Fig 1. Experimental pipeline for MLLM-assisted classroom behavior annotation. Bounding-box crops are annotated by two 7B-scale MLLMs, converted into pseudo-label sets through filtering strategies, and used to fine-tune CLIP with linear probing or LoRA. The same pipeline also yields annotation-quality and retention diagnostics.

Filtering Strategies

For implementation audit purposes, the pipeline records four filtering variants, but only two of them represent substantively different selection rules under the executed protocol:

- (1) **None (unfiltered):** All crops for which Qwen produces a valid index are retained. The pipeline uses the Qwen prediction as the pseudo-label.
- (2) **Valid-Output-Qwen:** Only crops where Qwen outputs a valid in-range index are retained (identical to None in practice, as Qwen has a near-100% valid-output rate).
- (3) **Valid-Output-LLaVA:** Only crops where LLaVA outputs a valid index are retained.
- (4) **Agreement:** Only crops where Qwen and LLaVA assign the same index are retained; crops with disagreement are excluded, and the shared index is used as the pseudo-label.

Because the annotation prompt enforces an index-only response and does not expose calibrated confidence scores, confidence thresholding is not available as a distinct scalar filter in the executed runs. In practice, Valid-Output-Qwen collapses to None and Valid-Output-LLaVA differs only through valid-output screening, so the downstream analysis focuses on None and Agreement as the two informative representative strategies. The goal of this comparison is therefore not to optimise a confidence threshold, but to test whether the readily available screening signals produced by the executed annotation protocol—valid-output screening and cross-model agreement—improve over unfiltered pseudo-labels under the same prompt and crop setting.

CLIP Fine-Tuning

Backbone

CLIP ViT-L/14 with OpenAI pretrained weights [13] serves as the visual backbone throughout, matching the model family used in prior zero-shot evaluation on these datasets [33]. The implementation uses `open_clip_torch` version 2.32.0. The image encoder produces 768-dimensional embeddings.

Fine-Tuning Methods

Two parameter-efficient fine-tuning strategies are compared:

Linear Probe (LP): The CLIP visual encoder is fully frozen. A single linear classification head ($768 \rightarrow C$, where C is the number of classes) is trained on the pseudo-labeled crops using cross-entropy loss.

Low-Rank Adaptation (LoRA): LoRA layers [34] with rank $r = 8$ and scaling factor $\alpha = 16$ are injected into the two MLP layers (`mlp.c_fc` and `mlp.c_proj`) of all transformer blocks in the visual encoder. LoRA layers are injected only into the MLP projections, as replacing the `MultiheadAttention` output projection in the `open_clip` implementation interfered with direct access to the underlying `out_proj.weight` and `out_proj.bias` attributes. This introduces 1.97M LoRA parameters while keeping the remaining ~ 303 M CLIP parameters frozen. The linear classification head is trained jointly with the LoRA parameters.

Training Details

All models are trained for 20 epochs using AdamW optimiser ($\text{lr} = 10^{-4}$, weight decay = 0.01) with cosine annealing learning rate schedule. The reported main-matrix and targeted-replication runs use batch size 128, distributed across two NVIDIA A100-SXM4-40GB GPUs via `DataParallel`. No gradient accumulation is used, so the effective optimisation batch size equals the per-step global batch size specified in each run. Best model selection is based on validation accuracy. For the main downstream matrix, LP conditions are averaged over repeated seeds (all datasets: five seeds, 43–47). A targeted LoRA replication over five seeds (43–47) is reported for BowTurnHead and TeacherBehavior under None and GT; the remaining LoRA values stay as protocol-matched single-run evaluations (seed 42). Retention-ratio and selective-routing diagnostics are reported as fixed-seed runs (seed 42). An overview of seed usage by module is provided in Table 4. Pseudo-label generation and CLIP fine-tuning remain split-isolated: validation crops are used only for annotation evaluation and model selection, whereas pseudo-label files consumed by CLIP fine-tuning are generated exclusively from the training split.

Experimental Conditions

For each of the three sub-datasets and two fine-tuning methods, the study evaluates three pseudo-label sources:

- **None:** Unfiltered MLLM pseudo-labels (Qwen predictions).
- **Agreement:** Agreement-filtered pseudo-labels.
- **GT:** Ground-truth human annotations (supervised reference).

This yields $3 \times 2 \times 3 = 18$ experimental conditions in total. The zero-shot CLIP baseline (no fine-tuning, prompt-engineered text embeddings) provides the primary practical reference. Prompt-adaptation methods such as CoOp and CoCoOp are not included as direct baselines here because they require target-domain optimisation; the present comparison isolates the value of pseudo-label supervision against a training-free reference and a ground-truth upper bound under the same crop protocol. The main-result zero-shot values (Table 8) are obtained under the same bbox-crop validation protocol using the canonical zero-shot evaluation from `step0b_cape_zeroshot.py`. Per-class prompt-ensemble outputs from the same evaluation pipeline are used separately in the anchor-proxy analysis and the selective-routing experiment.

Table 4. Seed policy by analysis module. Main-matrix LP uses repeated-seed evaluation; targeted LoRA replication is reported for BowTurnHead and TeacherBehavior under None and GT; auxiliary diagnostics and the remaining LoRA reference cells remain protocol-matched single-run/fixed-seed analyses in this revision.

Module	Protocol scope	Seed policy
Main downstream matrix (LP)	3 datasets $\times$ 3 label sources	5 seeds (43–47)
Targeted LoRA replication	BowTurnHead + TeacherBehavior; None/GT	5 seeds (43–47)
Main downstream matrix (remaining LoRA)	HandriseReadWrite + agreement conditions	single run (42)
Retention-ratio curves	3 datasets, LP diagnostics	fixed seed (42)
Selective routing (Scheme B)	TeacherBehavior oracle diagnostic	fixed seed (42)
Cross-model annotation check	5 MLLMs annotation quality	deterministic eval protocol

Integrated Diagnostic Analyses

Beyond the main annotation-quality and downstream fine-tuning comparisons, the study design includes four linked diagnostic analyses. Each addresses a different decision point in the pseudo-labeling workflow: whether categories are visually grounded enough for automatic annotation, whether agreement filtering fails because of retention loss or sampling bias, whether category-aware routing can serve as an upper-bound diagnostic on the hardest taxonomy, and whether the main annotation patterns remain stable under model substitution. Together, these analyses are intended to explain the main findings, not to replace repeated-seed downstream validation.

Selective Annotation (Scheme B)

To validate the diagnostic utility of visual anchoring as an annotation routing criterion, the study runs a selective annotation experiment on TeacherBehavior. The four high-anchor categories—*blackboard-writing*, *teacher*, *screen*, and *blackboard*—are trained using unfiltered pseudo-labels (None strategy) with a linear probe classification head restricted to those four classes. This selective routing is evaluated only as an oracle upper-bound diagnostic rather than as a directly deployable pipeline. The four low-anchor categories—*guide*, *answer*, *on-stage interaction*, and *stand*—are not included in fine-tuning; at inference time, these categories are handled by the frozen zero-shot CLIP classifier built with a prompt ensemble. During offline evaluation, the final prediction for each validation crop selects the fine-tuned head for high-anchor inputs and the zero-shot classifier for low-anchor inputs using the ground-truth category only as an oracle routing variable. The resulting overall 8-class accuracy should therefore be interpreted strictly as an upper-bound diagnostic for category-aware routing rather than as a directly deployable inference pipeline. A deployable variant would need a separate routing predictor derived from observable cues or proxy scores rather than ground-truth labels.

Pilot implementation tests with LoRA showed rapid overfitting and severe degradation of the low-anchor zero-shot branch, because LoRA modifies the shared visual encoder used by both branches. Scheme B reporting therefore focuses on linear probing.

Retention-Ratio Curve

To characterise the relationship between training data volume and downstream performance more precisely than the two-endpoint comparison afforded by the None versus Agreement conditions, the study evaluates intermediate retention ratios of 25%, 50%, and 75% on BowTurnHead and HandriseReadWrite. For each ratio, a random subset of the corresponding size is drawn from the unfiltered pseudo-label training set (fixed seed 42), and a linear probe is trained for 20 epochs under identical hyper-parameters. The resulting

five-point curves (25%, 50%, 75%, 100% from step3, and the Agreement endpoint) replace the two-endpoint figure in the original analysis. These retention-ratio diagnostics intentionally remain fixed-seed comparisons so that data-volume effects can be read against a single controlled subset draw; the main LP results in Table 8 instead report repeated-seed averages (all datasets: five seeds, 43–47).

Cross-Model Robustness Boundary Check

To test whether the main annotation findings depend strongly on the original two-model main pipeline (Qwen2-VL-7B and LLaVA-1.5-7B), the study includes a cross-model robustness check with three further open-weight MLLMs: Qwen2.5-VL-7B-Instruct, Qwen2.5-VL-32B-Instruct, and Gemma-3-27B-IT. The Gemma run loads the quantized Hugging Face checkpoint `unsloth/gemma-3-27b-it-bnb-4bit`, which implements the same Gemma-3-27B-IT model family used for this comparison. These runs use the same bbox-crop protocol, index-only prompt, and label decoding as the main annotation pipeline. Validation annotations are collected for all three sub-datasets, and TeacherBehavior is additionally processed on the training split to check whether the same category-level failure pattern persists outside the validation set. The cross-model analysis focuses on per-model pseudo-label accuracy and on pairwise agreement-subset accuracy among model pairs. Its purpose is not to redefine the main pipeline, but to test whether the main conclusions about model dependence and intent-dependent failure remain stable under model substitution. This robustness check therefore strengthens external validity with respect to annotator choice, but it does not by itself resolve the within-protocol variance question created by the remaining single-run downstream settings.

Operational Proxy for Visual Anchoring

To make visual anchoring operational in this study, the analysis defines a category-level proxy score using protocol-matched per-class zero-shot CLIP accuracy:

$$\text{AnchorProxy}(y) = \text{Acc}_y^{\text{ZS}}.$$

Here, Acc_y^{ZS} denotes the per-class accuracy of zero-shot CLIP (prompt-ensemble setting) under the same bbox-crop validation protocol used for all downstream experiments. Higher values indicate that category y is more visually anchored, whereas lower values indicate stronger dependence on intent or context. This proxy supports diagnostic interpretation and is not claimed as a causal estimator or as an independent ground-truth measure of visual anchoring. Because the proxy is derived from another classifier’s observed performance, it should be interpreted as an operational screening cue within this evaluation setup rather than as an independent explanatory measurement. In addition, because Scheme B routes low-anchor classes through the same zero-shot CLIP branch used to construct $\text{AnchorProxy}(y)$, these two signals are operationally coupled in this study and should not be interpreted as independent confirmatory evidence.

Results

Results are organised to answer the paper’s central question in stages. The section first establishes where MLLM pseudo-labels are reliable enough to serve as plausible training signal, using annotation-quality and per-category evidence. It then tests whether those pseudo-labels actually help downstream CLIP adaptation relative to zero-shot inference and the ground-truth reference. Finally, diagnostic and robustness analyses delimit why agreement filtering fails and how strongly the conclusions depend on annotator choice.

MLLM Annotation Quality

Table 5 reports pseudo-label accuracy and retention rates for the None and Agreement strategies on the validation split of each sub-dataset. Each retained bounding-box crop is treated as one evaluation instance, so both accuracy and retention are defined at bbox level rather than at whole-image level.

Table 5. MLLM pseudo-label annotation accuracy and retention rate on the validation split. “Pseudo-label acc.” = percentage of pseudo-labels matching the ground-truth label for retained bbox crops. Under the None strategy, “Retained” is the number of crops with a valid Qwen index; Qwen produces valid outputs for more than 99.9% of validation crops. These retention rates are computed on the validation split only; training-set retention rates used later in the retention-ratio analysis are reported separately in the text and figure.

Sub-dataset	None (Qwen)		Agreement (Qwen+LLaVA)		
	Pseudo-label acc. (%)	Retained	Pseudo-label acc. (%)	Retained	Ret. rate (%)
BowTurnHead	82.55	3,753	41.76	613	16.3
HandriseReadWrite	70.60	12,843	73.97	7,122	55.4
TeacherBehavior	41.21	15,327	47.57	10,167	66.3

Three observations follow. First, annotation accuracy decreases monotonically with task complexity: from 82.6% for the binary BowTurnHead to 41.2% for the eight-class TeacherBehavior. Second, Agreement filtering is not a uniformly reliable default: it improves retained-subset accuracy on HandriseReadWrite and TeacherBehavior, but substantially reduces both retention and accuracy on BowTurnHead. Only 16.3% of BowTurnHead validation crops survive the agreement filter, and their accuracy falls from 82.55% to 41.76%. Third, even after filtering, annotation accuracy remains well below the ground-truth reference for HandriseReadWrite and TeacherBehavior. To avoid ambiguity, the manuscript distinguishes these validation-set retention rates (16.3%, 55.4%, and 66.3%) from the training-set retention rates used in downstream fine-tuning (20.0%, 55.9%, and 66.5%).

Pairwise cross-model agreement follows the same pattern. Because both models produce valid outputs for almost every validation crop, the agreement rate is 16.33% on BowTurnHead, 55.46% on HandriseReadWrite, and 66.33% on TeacherBehavior, leaving corresponding disagreement rates of 83.67%, 44.54%, and 33.67%. BowTurnHead is the clearest failure case: the two models disagree on 3,140 of 3,753 validation crops, so the agreement filter keeps only a small and highly selective subset rather than a uniformly cleaner one.

Table 6. BowTurnHead validation-set confusion counts for Qwen and LLaVA. Rows denote ground-truth labels; columns denote model predictions.

GT class	Qwen→bow	Qwen→turn	LLaVA→bow	LLaVA→turn
bow-head	237	303	534	6
turn-head	352	2,861	3,193	20

Table 6 clarifies why agreement filtering collapses on BowTurnHead. Qwen predicts *turn-head* on most *turn-head* crops, whereas LLaVA collapses almost the entire *turn-head* class into *bow-head*. The dominant disagreement pattern is ground-truth *turn-head* with Qwen = *turn-head* and LLaVA = *bow-head* (2,842 crops), followed by ground-truth *bow-head* with Qwen = *turn-head* and LLaVA = *bow-head* (297 crops). Within the 613 agreed crops, the models jointly predict only 19 *turn-head* instances correctly but jointly mislabel 351 *turn-head* instances as *bow-head*, so agreement selects shared bias rather than shared correctness.

Per-Category Analysis of TeacherBehavior

Table 7 presents per-category annotation accuracy under the Agreement strategy for TeacherBehavior, revealing a clear bimodal distribution.

Table 7. Per-category MLLM annotation accuracy on TeacherBehavior (Agreement strategy, validation split). Accuracy is reported to two decimal places to match the other quantitative tables.

Category	Accuracy (%)	n
blackboard-writing	98.61	216
teacher	92.01	2,729
screen	69.41	1,154
blackboard	57.42	2,238
answer	0.00	415
guide	0.00	327
on-stage interaction	0.00	30
stand	0.85	3,058

The upper group comprises categories with *strong visual anchors*: each label corresponds to a distinctive, self-contained visual element (chalk marks on a blackboard; a projection screen; the blackboard surface itself). The lower group comprises categories whose labels are determined by *behavioural intent* rather than visual appearance: “answer” and “guide” are visually indistinguishable from a general lecture posture; “stand” is present in nearly every frame regardless of the actual activity. The exact 0.00% values for *answer*, *guide*, and *on-stage interaction* therefore indicate a systematic failure on intent-defined classes rather than a rounding artifact. An index-order error would be expected to induce a stable permutation pattern across all classes and pipeline stages, but the observed errors are class-specific and asymmetric instead. The error flows are also highly structured. Across all 853 validation instances of *answer*, Qwen maps 831 to *teacher*, while LLaVA splits the same class mainly between *teacher* (420) and *guide* (363); the agreement subset contains 415 crops and none is correct. Across 449 *guide* instances, Qwen predicts *teacher* 444 times and LLaVA predicts *teacher* 330 times, again yielding 0.00% agreement accuracy. For *on-stage interaction*, Qwen predicts *teacher* on 130 of 149 crops, whereas LLaVA predicts *on-stage interaction* on 84 of 149 crops but rarely agrees with Qwen; only 30 crops enter the agreement subset and all 30 are wrong. These patterns support a semantic-confusion explanation rather than an indexing error. The BowTurnHead confusion counts in Table 6 and the class-specific error flows reported here provide the direct audit evidence for that interpretation. This category-level distinction is referred to throughout the analysis as *visual anchoring*.

Fig 2 visualizes this separation. Categories with strong, object-like visual anchors occupy the high-accuracy group, whereas intent-dependent teacher behaviors form a near-zero-accuracy failure group.

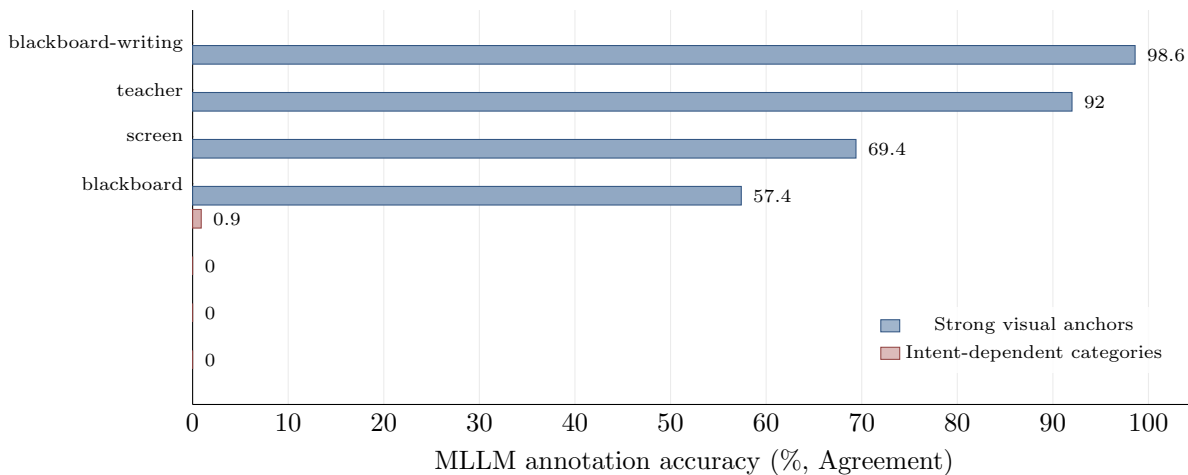


Fig 2. Per-category MLLM annotation reliability on TeacherBehavior under the Agreement strategy. Categories with explicit visual anchors are annotated reliably, whereas categories requiring intent inference drop to near-zero accuracy in the reported setting.

Representative crops in Fig 3 make the same contrast concrete. Classes such as *blackboard-writing* and

screen are tied to localized objects or surfaces that remain visually specific within one crop, whereas *answer* and *guide* reduce to generic teaching or teacher-student interaction postures despite carrying different semantic labels.

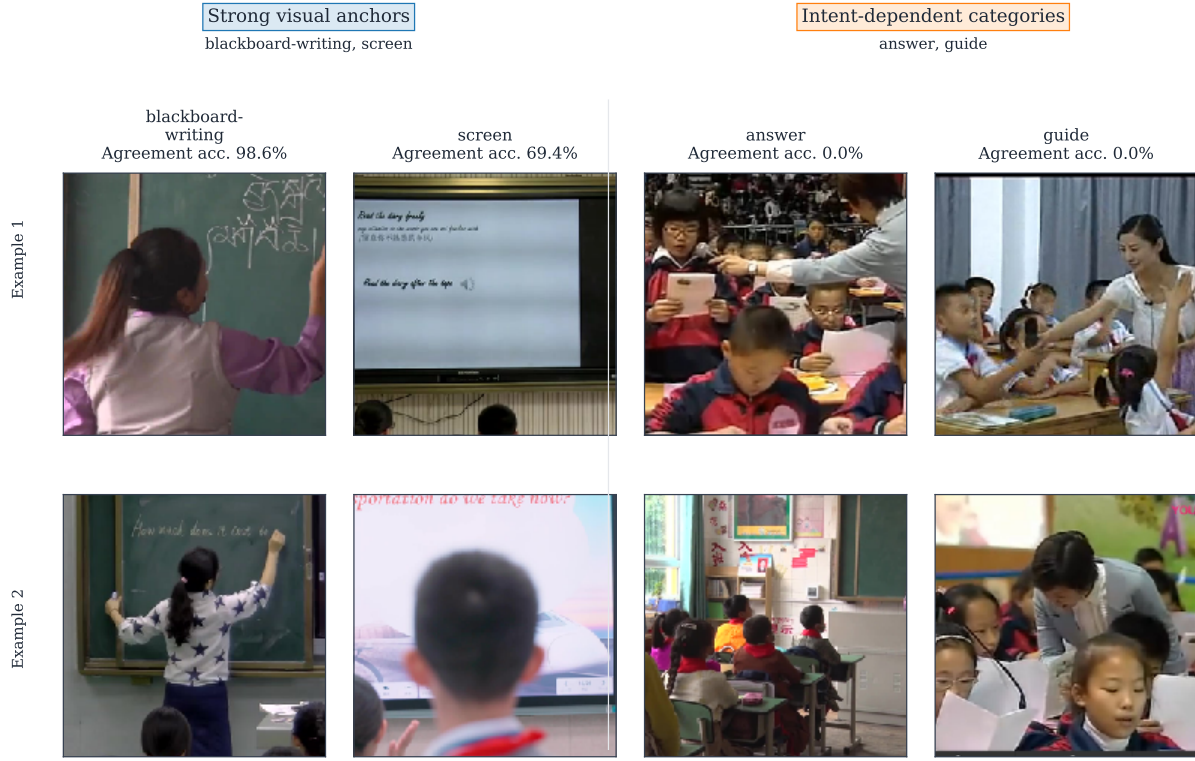


Fig 3. Representative crops from the TeacherBehavior validation agreement subset. High-anchor classes are associated with localized visual cues, whereas low-anchor intent-defined classes remain visually close to generic classroom interaction. The agreement accuracies shown above each category match Table 7.

Fine-Tuning Results

Table 8 presents the complete $3 \times 3 \times 2$ result matrix together with the zero-shot CLIP baseline.

Table 8. Validation accuracy (%) of CLIP ViT-L/14 under all experimental conditions. LP values report mean $\pm$ SD over repeated seeds (all datasets: five seeds, 43–47); LoRA values report mean $\pm$ SD for the targeted five-seed replication conditions (BowTurnHead and TeacherBehavior under None and GT; seeds 43–47) and seed-42 protocol-matched references elsewhere. **Bold** = best result per dataset. ZS = zero-shot CLIP baseline (no fine-tuning). LP = linear probe. LoRA = low-rank adaptation. The repeated-seed LoRA cells therefore correspond only to those four replicated conditions; see Finding 5 for the associated curve analysis.

Dataset	ZS	None		Agreement		GT	
		LP	LoRA	LP	LoRA	LP	LoRA
BowTurnHead	42.37	88.02 $\pm$ 0.12	88.65 $\pm$ 0.52	15.53 $\pm$ 1.00	19.69	93.35 $\pm$ 0.08	97.91$\pm$0.10
HandriseReadWrite	56.88	75.64 $\pm$ 0.13	76.69	52.88 $\pm$ 0.08	59.25	79.43 $\pm$ 0.20	87.06
TeacherBehavior	37.13	42.38 $\pm$ 0.07	43.22 $\pm$ 0.46	43.01 $\pm$ 0.05	45.18	69.56 $\pm$ 0.05	74.71$\pm$0.11

Fig 4 visualizes the same result matrix while keeping evidence levels explicit. The figure highlights three trends that are less immediately visible from the table alone: unfiltered pseudo-labels substantially improve

over zero-shot on BowTurnHead and HandriseReadWrite, agreement filtering is consistently inferior to unfiltered pseudo-labels except for a marginal TeacherBehavior LP gain of 0.63 pp, and ground-truth supervision remains the upper reference across all datasets.

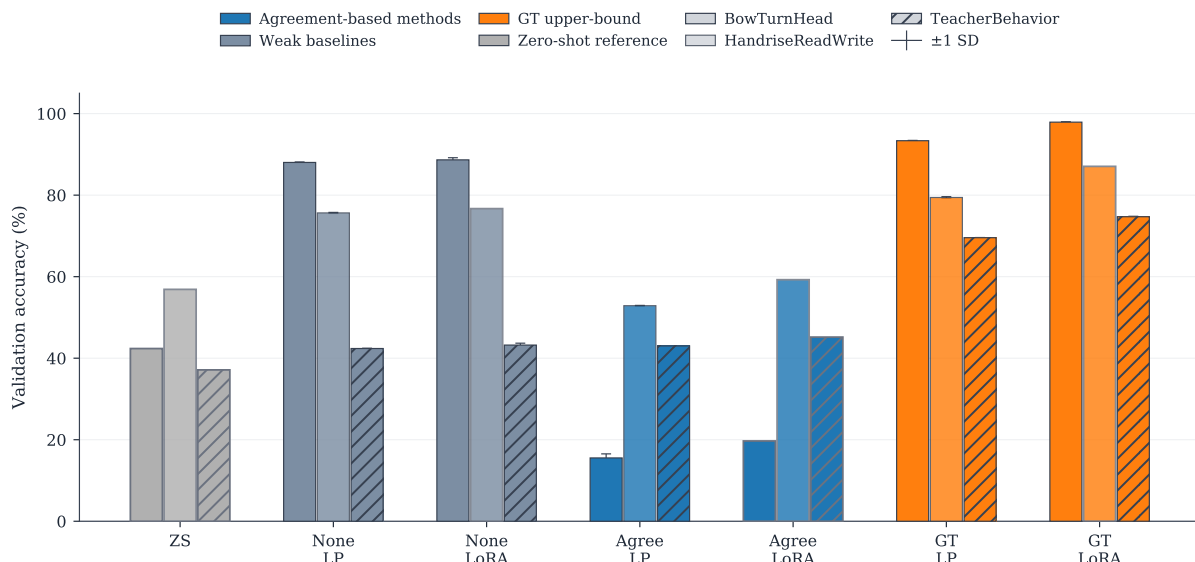


Fig 4. Overview of the full evaluation matrix. The plot includes the three zero-shot baselines and all 18 CLIP fine-tuning conditions formed by three datasets, three label sources, and two fine-tuning methods. Solid bars denote repeated or replicated conditions; hatched bars denote single-run references; error bars show ± 1 SD when available. ZS = zero-shot CLIP; LP = linear probing; LoRA = low-rank adaptation; Agree = agreement-filtered pseudo-labels. Here, ZS bars follow the baseline zero-shot setting used in Table 8 (step0b_cape_zeroshot.py); LP bars use repeated-seed means, whereas LoRA bars combine the targeted five-seed replication values for BowTurnHead and TeacherBehavior under None and GT with protocol-matched seed-42 references elsewhere. Prompt-ensemble zero-shot values are reported separately in the selective-analysis section.

Cross-Model Robustness as a Boundary Check

After the main downstream comparison, Table 9 summarises validation-set annotation accuracy for the five tested MLLMs under the common bbox-crop and index-only prompting protocol. The purpose of this analysis is not to redefine the main pipeline, but to test whether the category-level reliability pattern persists beyond the original two-model pair. The comparison shows that no single annotator is uniformly best across datasets: Qwen2.5-7B performs best on BowTurnHead, Qwen2 performs best on HandriseReadWrite, and Qwen2.5-7B and Gemma-3-27B are nearly tied on TeacherBehavior. At the pairwise level, agreement-subset accuracy varies substantially with the model pair, ranging from 22.41% to 89.58% on BowTurnHead, 57.60% to 78.14% on HandriseReadWrite, and 40.43% to 49.47% on TeacherBehavior. This pattern indicates that agreement quality is model-pair dependent rather than a model-independent denoising effect.

Table 9. Five-model cross-model robustness check of validation-set annotation accuracy under a common bbox-crop and index-only prompting protocol. Qwen2 = Qwen2-VL-7B-Instruct; LLaVA = LLaVA-1.5-7B; Qwen2.5-7B = Qwen2.5-VL-7B-Instruct; Qwen2.5-32B = Qwen2.5-VL-32B-Instruct; Gemma-3-27B = Gemma-3-27B-IT. Values are bbox-level pseudo-label accuracy (%).

Sub-dataset	Qwen2	LLaVA	Qwen2.5-7B	Qwen2.5-32B	Gemma-3-27B	Best
BowTurnHead	82.55	14.76	84.86	63.78	75.19	Qwen2.5-7B
HandriseReadWrite	70.59	53.20	51.74	69.11	57.21	Qwen2
TeacherBehavior	41.21	37.39	44.86	36.69	44.65	Qwen2.5-7B / Gemma-3-27B

Table 10 isolates the two clearest intent-dependent failure classes in TeacherBehavior. The *answer* category remains near-failure across all five models (0.00%–5.16%), and *guide* remains very weak overall (0.00%–24.05%), with four of the five models below 1%. The exception is LLaVA-1.5-7B (24.05% on *guide*), which is more consistent with a model-specific decision bias than robust intent-cue recognition, because the same model achieves only 14.76% overall on BowTurnHead. The same near-collapse therefore persists under substantial model substitution, strengthening the interpretation that these labels are difficult to resolve from single-frame crops under the present task setup.

Table 10. Cross-model validation on the two clearest intent-dependent TeacherBehavior categories. Values are validation-set bbox-level accuracy (%). n = number of validation crops for the category.

Category	n	Qwen2	LLaVA	Qwen2.5-7B	Qwen2.5-32B	Gemma-3-27B
guide	449	0.22	24.05	0.00	0.89	0.22
answer	853	0.00	0.00	1.17	5.16	1.41

Discussion

Finding 1: Agreement Filtering Fails Through Data-Volume Collapse, Distributional Bias, and Quality Ceiling

The central lesson of Table 5, Table 8, and Fig 6 is not simply that agreement filtering can underperform, but that it fails through three distinct mechanisms. BowTurnHead shows a data-volume failure: the filter discards enough examples to push training below a viable retention threshold. HandriseReadWrite shows a different mechanism: at moderate retention, random subsampling remains close to the full-data baseline while the agreement subset falls well below it, indicating biased selection rather than mere data reduction. TeacherBehavior shows a third regime: once pseudo-label noise becomes the dominant bottleneck, increasing retention changes little, so filtering neither rescues the labels nor raises the attainable ceiling in a meaningful way.

Fig 5 shows that the agreement subset does not preserve the original training distribution. It over-selects *bow-head* in BowTurnHead (35.76% to 63.40%), shifts HandriseReadWrite toward *hand-raise* at the expense of *read* (30.52% to 43.57%; 50.80% to 36.50%), and in TeacherBehavior over-selects *teacher* while under-selecting *stand* (21.03% to 26.45%; 34.51% to 31.32%). The filter therefore changes not only data volume but also the class mix seen during adaptation.

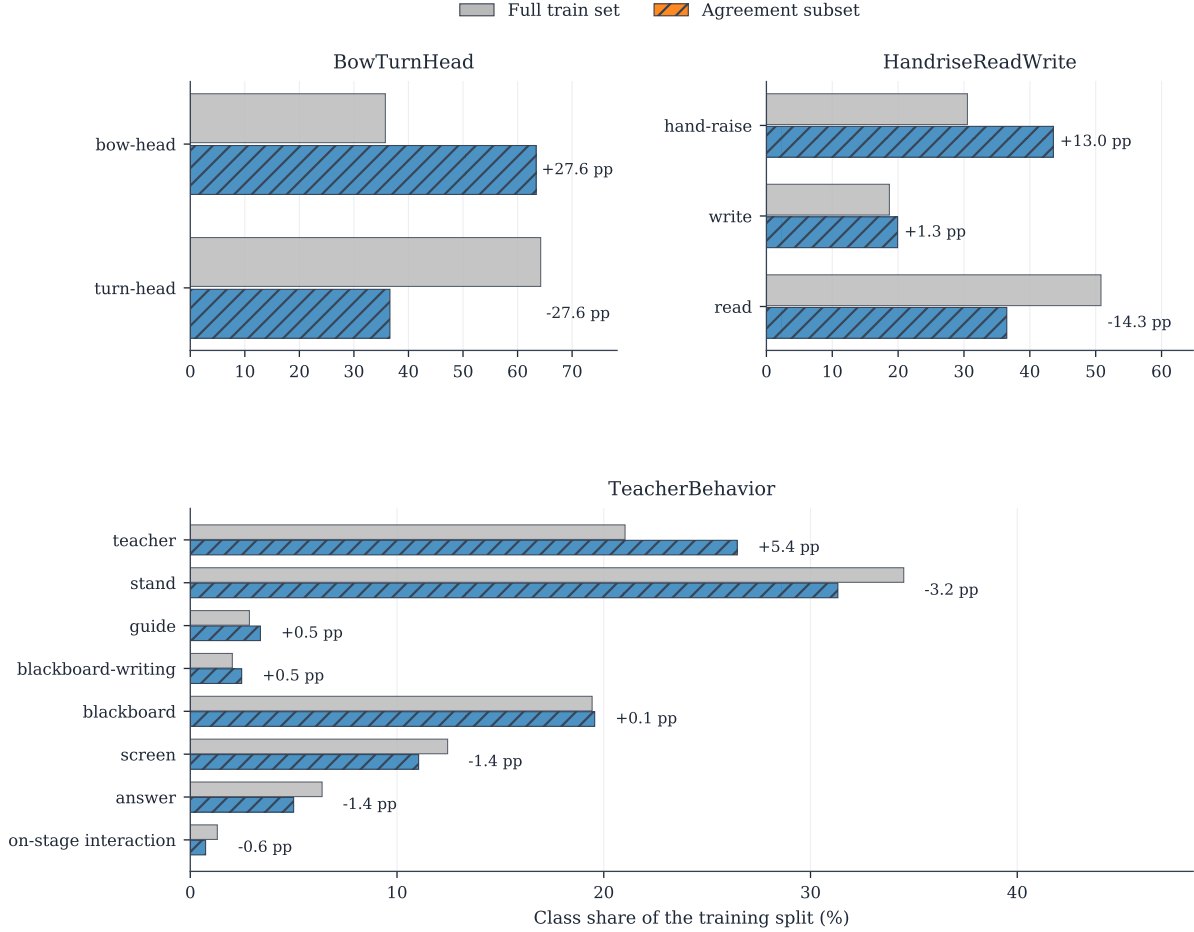


Fig 5. Class-composition shift between the full training split and the agreement subset for each dataset. Labels on the agreement bars report percentage-point change relative to the full training distribution, making the direction and magnitude of selection bias explicit.

These three regimes support a stronger claim than “agreement is imperfect.” Agreement is not a portable denoising rule in this setting because it can simultaneously alter data volume, class composition, and sample difficulty. The five-seed LP averages remain the primary downstream evidence, whereas the retention curves serve as mechanism checks. Taken together, they imply that agreement-filtered subsets should be treated as audited special cases, not as default pseudo-label training data. The added cross-model comparison points in the same direction: agreement quality varies materially with the model pair, so simple annotator substitution does not turn the heuristic into a stable strategy.

Finding 2: Visual Anchoring Explains When Agreement Filtering Fails

The most useful interpretation of Table 7 is as a mechanism-level explanation for the failure patterns above. Annotation reliability depends less on nominal model scale than on whether a class is visually observable in a single crop. Classes such as *blackboard-writing* and *teacher* expose distinctive visual cues that MLLMs can align with pretraining knowledge, whereas categories such as *answer*, *guide*, and *on-stage interaction* depend more on pedagogical intent and discourse context than on a unique frame-level signature. The protocol-matched zero-shot CLIP results are therefore used here as an operational anchoring proxy, not as an independent causal explanation: $\text{AnchorProxy}(y) = \text{Acc}_y^{\text{ZS}}$ helps distinguish categories that are visually legible under the current protocol from those that are not.

The five-model boundary check strengthens this interpretation. Although the best annotator changes

across datasets, the same intent-dependent failure classes remain near-collapse under model substitution. That consistency suggests a task-level bottleneck rather than a defect specific to one annotator pair. In practical terms, the limiting issue is single-frame observability: if the label cannot be recovered from an isolated crop, changing the annotator is unlikely to solve the problem. Under this reading, visual anchoring is best treated as a screening diagnostic for when pseudo-label supervision is plausible. High-anchor categories are plausible candidates for pseudo-label supervision, whereas low-anchor categories should default to stronger prompting, added audit, human annotation, or richer temporal and multimodal evidence.

More broadly, these findings challenge a common assumption in pseudo-label pipelines: that inter-model agreement is a reliable proxy for label quality. In the present setting, agreement preferentially retains samples on which both models express the same perceptual bias, rather than samples that are objectively easier to classify. When two annotators inherit similar pretraining corpora or architectural inductive biases, their agreement can therefore reflect shared blind spots rather than genuine label confidence. Under that view, agreement filtering is better understood not as a denoising operation, but as a selection mechanism whose behaviour depends on the model pair and on the visual discriminability of the target categories.

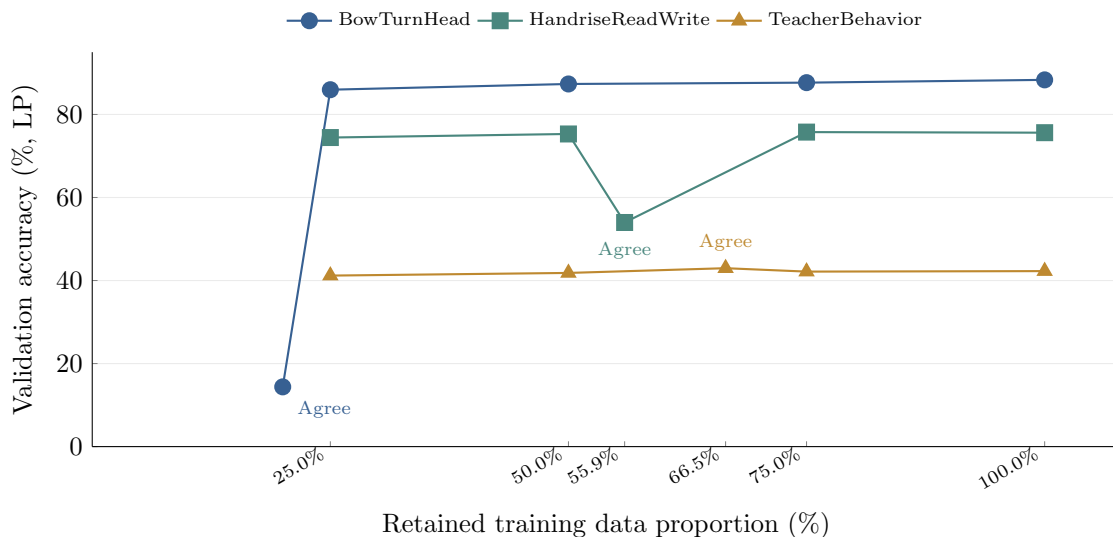


Fig 6. Retention-ratio curves for BowTurnHead, HandriseReadWrite, and TeacherBehavior under linear probing with unfiltered pseudo-labels (None strategy), augmented with the agreement-filtered endpoint (labelled “Agree”). All plotted points are fixed-seed diagnostic runs (seed 42), whereas Table 8 reports repeated-seed LP means for the main downstream comparison. The x-axis reports *training-set* retention rates, whereas Table 5 reports *validation-set* retention rates for annotation quality. Random subsampling remains stable above the critical retention threshold for all three datasets. The agreement-selected endpoint falls far below the random-retention trend for BowTurnHead and HandriseReadWrite, whereas TeacherBehavior shows only a small positive deviation.

Finding 3: Anchor-Aware Selective Annotation Provides an Informative Upper-Bound Diagnostic

The selective-annotation experiment addresses a narrower question than the main matrix: if some categories are visually anchorable and others are not, is uniform pseudo-label treatment the wrong abstraction? Table 11 suggests that it is. Routing high-anchor categories through pseudo-label supervision while leaving low-anchor categories to the zero-shot branch improves the TeacherBehavior score over both full pseudo-label fine-tuning and the zero-shot baseline. The result should still be read as a boundary diagnostic rather than a deployable method. Scheme B uses oracle routing at evaluation time, and the current AnchorProxy is operationally coupled to the zero-shot branch used for low-anchor classes. Its value is therefore explanatory: it supports the claim that pseudo-label supervision is most defensible when applied selectively to categories with sufficient visual evidence.

Finding 4: Task Complexity Moderates the Benefit of Pseudo-Label Fine-Tuning

The downstream comparison answers the central question conditionally. Unfiltered pseudo-labels can beat zero-shot CLIP when the task is visually grounded enough that label noise does not erase the training signal. That is the pattern on BowTurnHead and, more moderately, on HandriseReadWrite. The same strategy is far less persuasive on TeacherBehavior, where the label semantics depend more heavily on intent and interaction context.

Ground-truth fine-tuning clarifies the distinction. Clean labels remain strongly useful across all three datasets, so the weak pseudo-label result on TeacherBehavior should not be read as evidence that the task itself is resistant to adaptation. It is instead evidence that task complexity and label semantics jointly determine whether pseudo-label supervision is worth using. A practical decision rule follows: pseudo-label fine-tuning is most plausible when classes are visually grounded and semantically stable; as intent dependence increases, selective human annotation becomes more valuable.

Finding 5: The Attainable Performance Ceiling Is More Strongly Locked by Label Quality Than by Optimisation Path

The five-seed LoRA replications sharpen the same conclusion from a different angle. What changes most across conditions is not only mean accuracy but the ceiling itself. Ground-truth supervision produces both higher validation scores and much lower cross-seed variance, whereas pseudo-label supervision keeps the model in a noisier regime. Across the replicated BowTurnHead and TeacherBehavior settings, the mean cross-seed SD is 0.107% under GT supervision versus 0.495% under pseudo-label supervision. On TeacherBehavior in particular, the pseudo-label setting can fit the training set almost perfectly while validation remains trapped in a narrow band. That pattern is more consistent with memorising inconsistent decision boundaries than with simply choosing the wrong checkpoint or optimisation schedule.

This matters because it limits one easy counterargument to the main results. The weak pseudo-label outcome cannot be explained away by saying that LoRA merely needed more careful tuning. Under the present single-frame protocol, optimisation changes the training trajectory more than it changes the attainable validation regime. The remaining single-run LoRA cells should still be read cautiously, but the replicated seeds are sufficient to support a narrower conclusion: label quality constrains both performance and stability more strongly than optimisation path in the settings where the paper makes its strongest claims.

TeacherBehavior Case Study: What the Selective-Routing Gain Does and Does Not Show

The TeacherBehavior case study makes the boundary condition concrete. When agreement-based selection is unreliable, category-aware routing can still be informative as an upper-bound diagnostic, even though the present Scheme B implementation is not directly deployable. Table 11 summarises the result.

Table 11. Comparison of annotation strategies on TeacherBehavior (validation accuracy, %). The selective strategy uses pseudo-labels for high-anchor classes and zero-shot CLIP for low-anchor classes. High-anchor: blackboard-writing, teacher, screen, blackboard. Low-anchor: guide, answer, on-stage interaction, stand. The zero-shot baseline in this table uses the prompt-ensemble setting. **Selective routing uses ground-truth category as oracle routing variable at evaluation time and represents a diagnostic upper bound, not a deployable pipeline.**

Strategy	Overall (8-cl)	High-anchor (4-cl)	Low-anchor (4-cl)
Zero-shot CLIP (prompt ensemble)	53.43	—	—
None + LP (full pseudo)	42.38	—	—
Selective + LP	58.37	72.74	38.44
GT + LP (upper bound)	69.56	—	—

The gain comes from the high-anchor portion of the taxonomy, while the low-anchor classes remain better handled by the zero-shot branch. That is precisely why the result should not be interpreted as a new universal pipeline. It depends on oracle routing and on a fine-tuning setup that preserves the zero-shot branch.

This constraint also explains why LoRA is a poor fit for Scheme B. Once the shared encoder is altered, the fallback branch for low-anchor categories is no longer stable. The case study is therefore most useful as a design implication. If a taxonomy mixes visually explicit and intent-dependent classes, selective routing may be worth exploring, but only with an explicit routing mechanism and without presenting the oracle result as deployment-ready.

Limitations

Several limitations should be noted. First, the main annotation pipeline evaluates two 7B-scale locally deployable MLLMs, and the broader robustness analysis spans five models in total: Qwen2-VL-7B, LLaVA-1.5-7B, Qwen2.5-VL-7B-Instruct, Qwen2.5-VL-32B-Instruct, and Gemma-3-27B-IT. This broader evidence reduces concern that the reported failure patterns are unique to one particular model pair, but broader validation across more datasets, prompt regimes, and multimodal model families remains necessary. Whether larger-scale or video-aware models materially improve annotation reliability for intent-dependent categories therefore remains an open question. Second, the annotation pipeline uses image-level prompts without chain-of-thought reasoning or few-shot examples; more sophisticated prompting strategies may improve accuracy on intent-dependent categories. In particular, the index-only output constraint used here standardises parsing but does not expose uncertainty, so prompt sensitivity and abstention-aware prompting remain open questions. Third, the study uses a single benchmark (SCB) from Chinese K-12 classrooms; generalisability to other educational contexts and cultural settings requires further investigation. Fourth, the selective annotation experiment evaluates only linear probing; LoRA is incompatible with Scheme B due to encoder-level interference with the zero-shot classifier, and alternative architectures that decouple the fine-tuned head from the shared encoder (e.g., adapter layers, separate encoder branches) are left for future work. Fifth, the main LP conditions are averaged over five random seeds (all datasets: 43–47), but the LoRA replication now covers BowTurnHead and TeacherBehavior under None and GT across five seeds (43–47). However, the remaining LoRA settings and several diagnostic analyses remain single-run protocol-matched evaluations. The reported differences should therefore still be interpreted as controlled diagnostic comparisons rather than as full estimates of training variance across all conditions. The study therefore does not support formal significance claims for small differences in conditions that remain single-run, and future work should extend repeated-seed evaluation to the remaining LoRA and auxiliary diagnostic analyses, together with bootstrap- or permutation-based uncertainty estimates. The five-model robustness analysis partly strengthens external validity with respect to annotator choice, but it does not remove this internal validity limitation in the downstream fine-tuning matrix. The fine-tuning hyper-parameters are also held fixed across the main comparisons, so systematic learning-rate sensitivity is not evaluated in this study. Sixth, the present study treats visual anchoring as a diagnostic concept that is operationalised here with a task-specific proxy ($\text{AnchorProxy}(y) = \text{Acc}_y^{\text{ZS}}$), rather than a universal quantitative variable. Because this proxy is itself derived from classifier performance, it does not provide an independent measurement of anchoring and should be interpreted as an operational diagnostic rather than a standalone explanatory construct. Future work should evaluate more general category-level indicators and test whether their association with pseudo-label accuracy is stable across datasets and model families. Seventh, practical training-time baselines beyond the present matrix are not yet included (e.g., confidence-threshold pseudo-labeling, teacher-student/self-training variants, or prompt-adaptation methods such as CoOp/CoCoOp under matched supervision constraints). Although the current zero-shot and GT references bound performance, this omission limits engineering-side comparability and should be addressed in future extensions. Eighth, although the retention-ratio analysis and distribution-shift metrics (total-variation distance, KL divergence) provide partial controls for data-volume and sampling-bias effects, the three failure mechanisms identified for agreement filtering—data-volume collapse, distributional bias, and quality ceiling—are not fully causally isolated in the current experimental design. Agreement-filtered subsets differ from random subsets simultaneously in data volume, class distribution, sample difficulty, and visual diversity,

so the present mechanism attribution should be read as diagnostically motivated rather than causally verified. Controlled experiments that independently vary these factors—for example, class-balanced resampling matched to the agreement subset distribution, or difficulty-stratified retention curves—are left for future work.

Reproducibility Statement

All experiments are executed using the following pipeline scripts: zero-shot evaluation (baseline and prompt-ensemble settings), MLLM annotation (`step1_llm_annotate.py`), filtering analysis (`step2_filter_analysis.py`), CLIP fine-tuning (`step3_clip_finetune.py`), selective annotation (`step4_selective_annotation.py`), and retention-ratio curve (`step5_retention_curve.py`). This study reports dataset splits, model variants, prompt format, training hyper-parameters; LP main-matrix results are averaged over repeated seeds (all datasets: 43–47); targeted LoRA replications for BowTurnHead and TeacherBehavior under None and GT use seeds 43–47, while remaining LoRA and selected diagnostics are single-seed/fixed-seed evaluations (seed 42). The SCB dataset is publicly available [33]. The code repository and run configurations used for this study are publicly available at <https://github.com/zhanglizhuo/llm-annotation> (accessed on 27 April 2026), including the pipeline scripts, archived shell entry points, and result directories referenced in this paper. For reproducibility review, the repository should remain publicly accessible during the editorial process and include environment specifications (Python and PyTorch/CUDA versions), dependency installation instructions, and one-command run examples for each pipeline stage.

Conclusions

This study asked a practical question: when can MLLM-generated pseudo-labels improve classroom behavior recognition, and when does agreement filtering make them worse instead of better? Within the present SCB benchmark, the answer is conditional rather than global. Pseudo-labels help when the target classes are visually anchored enough that single-frame crops preserve the evidence needed for labeling. They do not help reliably when the taxonomy is dominated by intent-dependent classes, and cross-model agreement is not a safe remedy because it can remove training volume, distort class composition, and preserve shared bias instead of correctness.

The evidence supporting that answer is consistent across the manuscript. Category-level annotation results show a sharp divide between visually anchored classes and intent-dependent classes. The repeated-seed LP matrix shows that unfiltered pseudo-labels can outperform zero-shot CLIP on visually grounded datasets, whereas the clearest counterexample appears on BowTurnHead, where Agreement+LP falls to $15.53\% \pm 1.00\%$ while None+LP reaches $88.02\% \pm 0.12\%$. The targeted LoRA replications indicate that label quality constrains not only attainable accuracy but also run-to-run stability. The anchor-aware Scheme B analysis and the five-model robustness check then bound the claim: pseudo-label supervision is most defensible as a selective diagnostic tool for high-anchor categories, not as a universal replacement for human annotation or temporal and multimodal evidence.

The practical implication is therefore not “filter harder” but “screen the task first.” For SCB-like single-frame classroom sensing, a conservative workflow is to audit category anchoring before committing annotation budget, avoid agreement filtering as a default denoising rule, use pseudo-label supervision mainly on high-anchor categories, and reserve low-anchor categories for stronger zero-shot prompts, temporal or multimodal sensing, or human labelling. Under this interpretation, the contribution of the paper is a bounded answer to when MLLM pseudo-labeling is useful in classroom behavior recognition, together with the main failure modes that make it unreliable in the present setting.

Author contributions

Conceptualization, Y.M. and L.Z.; methodology, Y.M. and L.Z.; software, L.Z.; validation, Y.M. and L.Z.; formal analysis, Y.M.; investigation, Y.M.; data curation, Y.M. and L.Z.; visualization, L.Z.;

writing—original draft preparation, Y.M.; writing—review and editing, Y.M. and L.Z.; supervision, L.Z.

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Competing interests

The authors declare that no competing interests exist. The funders had no role in the design of the study; in the collection, analyses, or interpretation of data; in the writing of the manuscript; or in the decision to publish the results.

Data availability

The three SCB sub-datasets used in this study, plus the excluded single-class Discuss set, are publicly available on Hugging Face at <https://huggingface.co/datasets/wintonYF/SCB-Dataset> (accessed on 27 April 2026). These datasets are externally released resources and are not owned by this study. The experiment code and archived run configurations are available at <https://github.com/zhanglizhuo/llm-annotation> (accessed on 27 April 2026). The repository was publicly accessible at the time of access.

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